



[This question paper contains 2 printed pages.]

Sr. No. of Question Paper : 1116

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Your Roll No.....

Unique Paper Code : 235304

Name of the Paper : III.3 – Algebra II

Name of the Course : B.Sc. (Hons.) Mathematics, Part II

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt any **two** parts from each question.
3. All questions are compulsory.

1. (a) Let $G = \mathbb{Q} \setminus \{-1\}$. Define $*$ on G by $a*b = a + b + ab$ for all $a, b \in G$. Prove that $(G, *)$ is an abelian group. (6)

(b) (i) Let G be a group and H be a non-empty subset of G . Prove that H is a subgroup of G iff $a.b^{-1} \in H$ for all $a, b \in H$. (3)

(ii) Prove that the group of positive rational numbers under multiplication is not cyclic. (3)

(c) Define centralizer of $a \in G$, where G is a group.

(i) Prove that $C(a)$ is a subgroup of G .

(ii) Let $G = GL(2, \mathbb{R})$. Find $C \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. (6)

2. (a) For every non-negative integer n , prove that $n\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$. Moreover, prove that every subgroup of $\mathbb{Z}$ is of the form $m\mathbb{Z}$ for some non-negative integer m . (6)

(b) Let G be a group. Let $a, b \in G$ such that $ab = ba$ and $|a| = m$ and $|b| = n$. If $\langle a \rangle \cap \langle b \rangle = \{e\}$. Prove that G has an element of order l.c.m. (m, n) . (6)

(c) Prove that the order of a cyclic group is equal to the order of its generator. (6)

3. (a) Let H and K be two subgroups of G . Prove that HK is a subgroup of G iff $HK = KH$. (6.5)
- (b) (i) Show that for $n \leq 3$, $Z(S_n) = \{e\}$. (3.5)
 (ii) Suppose G is an abelian group with odd number of elements. Show product of all elements of G is identity. (3)
- (c) State Lagrange's Theorem and prove with the help of example that converse of Lagrange's Theorem does not hold in general. (6.5)
4. (a) Prove that every permutation of a finite set can be written as product of disjoint cycles. (6.5)
- (b) (i) Let H be a subgroup of G and let $a, b \in G$. Prove that either $aH = bH$ or $aH \cap bH = \phi$. (3)
 (ii) Let N be a normal subgroup of a group G . If N is cyclic, prove that every subgroup of N is also normal in G . (3.5)
- (c) (i) Given that G is a cyclic group, prove that G/N is also cyclic where N is a subgroup of G . Also give an example to show that converse is not true. (3.5)
 (ii) If N is a normal subgroup of G and $|G/N| = m$, prove that $x^m \in N$ for all x in G . (3)
5. (a) Define automorphism and inner automorphism induced by an element 'a' of a group G . Prove that $\text{Inn}(G)$ is a normal subgroup of $\text{Aut } G$. (6)
- (b) Let ϕ be a homomorphism from a group G to a group $\bar{G}$. If K is a normal subgroup of $\bar{G}$, then prove that $\phi^{-1}(K)$ is a normal subgroup of G . (6)
- (c) Let $G = U(10)$. Find the left regular representation $\bar{G}$ of G and verify that $G \approx \bar{G}$. (6)
6. (a) If M and N are normal subgroups of G and $N \leq M$, prove that $(G/N)/(M/N) \approx G/M$. (6.5)
- (b) (i) Let $\phi: G \rightarrow \bar{G}$ be a homomorphism. Define $\text{Ker } \phi$ and prove that $\text{Ker } \phi$ is a normal subgroup of G . (3)
 (ii) Show $\phi: Z_{12} \rightarrow Z_{12}$ by $\phi(x) = 3x$ is a homomorphism and find $\text{Ker } \phi$. Also find $\phi^{-1}(6)$. (3.5)
- (c) Let k be a divisor of n . Prove that $Z_n / \langle k \rangle \approx Z_k$. (6.5)